



Averages with Unequal Group Sizes

Grades 9-12 • SAT / ACT mathematics foundations

Problem-Solving and Data Analysis • Compute a combined mean using group sizes.

Original focused skill practice for learners preparing for college-entrance mathematics. Includes models, 16 practice tasks, six explain-and-check tasks, a four-item exit ticket, and worked solutions. Not a full practice exam.

PRINT • TEACH • REVISIT



Plan a focused study session

Learning objective

Compute a combined mean using group sizes.

Readiness

Use after learners have studied the relevant algebra, geometry, or data concepts. Grade tags describe a typical preparation audience; readiness depends on prior coursework.

Suggested use

Session 1: 10 minutes on worked examples and 20–30 minutes on practice. Session 2: review errors, then use explain-and-check tasks. Finish with the independent exit ticket. Pause and reteach a prerequisite when the model is not yet understood.

What this resource covers

The named mathematics skill is useful within SAT and ACT preparation. The packet does not cover every exam domain, reproduce an official section, provide standardized timing, or predict a score.

Materials

Pencil, paper, and a calculator for checking when useful. Practice both symbolic reasoning and calculator verification. Follow each exam publisher's current policies for an actual test.

Student directions

Read the entire question, define quantities, show a method, and check the result. For a written explanation, cite the calculation or feature that supports your conclusion.



Two worked examples

Model 1. Group A has 6 scores with mean 67. Group B has 18 scores with mean 87. Find the combined mean. Explain why simply averaging the two means can fail.

Total score = $6 \times 67 + 18 \times 87 = 1968$. Divide by 24 scores: 82. The group sizes determine each mean's weight.

Model 2. Group A has 4 scores with mean 65. Group B has 12 scores with mean 85. Find the combined mean. Explain why simply averaging the two means can fail.

Total score = $4 \times 65 + 12 \times 85 = 1280$. Divide by 16 scores: 80. The group sizes determine each mean's weight.

Pause before looking at a solution: which relationship, definition, or constraint determines the first step?

Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. Group A has 6 scores with mean 66. Group B has 9 scores with mean 86. Find the combined mean. Explain why simply averaging the two means can fail.

2. Group A has 6 scores with mean 67. Group B has 24 scores with mean 87. Find the combined mean. Explain why simply averaging the two means can fail.

3. Group A has 10 scores with mean 65. Group B has 15 scores with mean 85. Find the combined mean. Explain why simply averaging the two means can fail.



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

4. Group A has 6 scores with mean 68. Group B has 24 scores with mean 88. Find the combined mean. Explain why simply averaging the two means can fail.

5. Group A has 18 scores with mean 64. Group B has 9 scores with mean 84. Find the combined mean. Explain why simply averaging the two means can fail.

6. Group A has 18 scores with mean 63. Group B has 27 scores with mean 83. Find the combined mean. Explain why simply averaging the two means can fail.



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

7. Group A has 4 scores with mean 63. Group B has 12 scores with mean 83. Find the combined mean. Explain why simply averaging the two means can fail.

8. Group A has 18 scores with mean 65. Group B has 27 scores with mean 85. Find the combined mean. Explain why simply averaging the two means can fail.

9. Group A has 12 scores with mean 68. Group B has 21 scores with mean 88. Find the combined mean. Explain why simply averaging the two means can fail.



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

10. Group A has 12 scores with mean 62. Group B has 9 scores with mean 82. Find the combined mean. Explain why simply averaging the two means can fail.

11. Group A has 14 scores with mean 63. Group B has 24 scores with mean 83. Find the combined mean. Explain why simply averaging the two means can fail.

12. Group A has 6 scores with mean 68. Group B has 12 scores with mean 88. Find the combined mean. Explain why simply averaging the two means can fail.



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

13. Group A has 4 scores with mean 68. Group B has 18 scores with mean 88. Find the combined mean. Explain why simply averaging the two means can fail.

14. Group A has 10 scores with mean 66. Group B has 27 scores with mean 86. Find the combined mean. Explain why simply averaging the two means can fail.

15. Group A has 18 scores with mean 64. Group B has 24 scores with mean 84. Find the combined mean. Explain why simply averaging the two means can fail.



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

16. Group A has 16 scores with mean 66. Group B has 27 scores with mean 86. Find the combined mean. Explain why simply averaging the two means can fail.



Explain and check • Transfer practice

Solve the task, then explain a check or limitation. Compare your reasoning with the worked key only after making your own attempt.

1. Group A has 14 scores with mean 67. Group B has 18 scores with mean 87. Find the combined mean. Explain why simply averaging the two means can fail.

2. Group A has 12 scores with mean 63. Group B has 6 scores with mean 83. Find the combined mean. Explain why simply averaging the two means can fail.

Explain and check • Transfer practice

Solve the task, then explain a check or limitation. Compare your reasoning with the worked key only after making your own attempt.

3. Group A has 16 scores with mean 67. Group B has 9 scores with mean 87. Find the combined mean. Explain why simply averaging the two means can fail.

4. Group A has 16 scores with mean 67. Group B has 27 scores with mean 87. Find the combined mean. Explain why simply averaging the two means can fail.

Explain and check • Transfer practice

Solve the task, then explain a check or limitation. Compare your reasoning with the worked key only after making your own attempt.

5. Group A has 14 scores with mean 67. Group B has 12 scores with mean 87. Find the combined mean. Explain why simply averaging the two means can fail.

6. Group A has 6 scores with mean 62. Group B has 18 scores with mean 82. Find the combined mean. Explain why simply averaging the two means can fail.

Exit ticket • Independent check

Complete independently. Show reasoning and any relevant restriction. This is a classroom learning check, not a standardized test score.

1. Group A has 4 scores with mean 64. Group B has 27 scores with mean 84. Find the combined mean. Explain why simply averaging the two means can fail.

2. Group A has 12 scores with mean 65. Group B has 27 scores with mean 85. Find the combined mean. Explain why simply averaging the two means can fail.



Exit ticket • Independent check

Complete independently. Show reasoning and any relevant restriction. This is a classroom learning check, not a standardized test score.

3. Group A has 18 scores with mean 65. Group B has 9 scores with mean 85. Find the combined mean. Explain why simply averaging the two means can fail.

4. Group A has 4 scores with mean 64. Group B has 6 scores with mean 84. Find the combined mean. Explain why simply averaging the two means can fail.



Practice worked solutions

1. 78

Total score = $6 \times 66 + 9 \times 86 = 1170$. Divide by 15 scores: 78. The group sizes determine each mean's weight.

2. 83

Total score = $6 \times 67 + 24 \times 87 = 2490$. Divide by 30 scores: 83. The group sizes determine each mean's weight.

3. 77

Total score = $10 \times 65 + 15 \times 85 = 1925$. Divide by 25 scores: 77. The group sizes determine each mean's weight.

4. 84

Total score = $6 \times 68 + 24 \times 88 = 2520$. Divide by 30 scores: 84. The group sizes determine each mean's weight.

Practice worked solutions

5. 212/3

Total score = $18 \times 64 + 9 \times 84 = 1908$. Divide by 27 scores: 212/3. The group sizes determine each mean's weight.

6. 75

Total score = $18 \times 63 + 27 \times 83 = 3375$. Divide by 45 scores: 75. The group sizes determine each mean's weight.

7. 78

Total score = $4 \times 63 + 12 \times 83 = 1248$. Divide by 16 scores: 78. The group sizes determine each mean's weight.

8. 77

Total score = $18 \times 65 + 27 \times 85 = 3465$. Divide by 45 scores: 77. The group sizes determine each mean's weight.

Practice worked solutions

9. 888/11

Total score = $12 \times 68 + 21 \times 88 = 2664$. Divide by 33 scores: 888/11. The group sizes determine each mean's weight.

10. 494/7

Total score = $12 \times 62 + 9 \times 82 = 1482$. Divide by 21 scores: 494/7. The group sizes determine each mean's weight.

11. 1437/19

Total score = $14 \times 63 + 24 \times 83 = 2874$. Divide by 38 scores: 1437/19. The group sizes determine each mean's weight.

12. 244/3

Total score = $6 \times 68 + 12 \times 88 = 1464$. Divide by 18 scores: 244/3. The group sizes determine each mean's weight.

Practice worked solutions

13. 928/11

Total score = $4 \times 68 + 18 \times 88 = 1856$. Divide by 22 scores: 928/11. The group sizes determine each mean's weight.

14. 2982/37

Total score = $10 \times 66 + 27 \times 86 = 2982$. Divide by 37 scores: 2982/37. The group sizes determine each mean's weight.

15. 528/7

Total score = $18 \times 64 + 24 \times 84 = 3168$. Divide by 42 scores: 528/7. The group sizes determine each mean's weight.

16. 3378/43

Total score = $16 \times 66 + 27 \times 86 = 3378$. Divide by 43 scores: 3378/43. The group sizes determine each mean's weight.

Transfer worked solutions

1. 313/4

Total score = $14 \times 67 + 18 \times 87 = 2504$. Divide by 32 scores: 313/4. The group sizes determine each mean's weight.

2. 209/3

Total score = $12 \times 63 + 6 \times 83 = 1254$. Divide by 18 scores: 209/3. The group sizes determine each mean's weight.

3. 371/5

Total score = $16 \times 67 + 9 \times 87 = 1855$. Divide by 25 scores: 371/5. The group sizes determine each mean's weight.

4. 3421/43

Total score = $16 \times 67 + 27 \times 87 = 3421$. Divide by 43 scores: 3421/43. The group sizes determine each mean's weight.

Transfer worked solutions

5. 991/13

Total score = $14 \times 67 + 12 \times 87 = 1982$. Divide by 26 scores: 991/13. The group sizes determine each mean's weight.

6. 77

Total score = $6 \times 62 + 18 \times 82 = 1848$. Divide by 24 scores: 77. The group sizes determine each mean's weight.

Exit-ticket worked solutions

1. 2524/31

Total score = $4 \times 64 + 27 \times 84 = 2524$. Divide by 31 scores: $2524/31$. The group sizes determine each mean's weight.

2. 1025/13

Total score = $12 \times 65 + 27 \times 85 = 3075$. Divide by 39 scores: $1025/13$. The group sizes determine each mean's weight.

3. 215/3

Total score = $18 \times 65 + 9 \times 85 = 1935$. Divide by 27 scores: $215/3$. The group sizes determine each mean's weight.

4. 76

Total score = $4 \times 64 + 6 \times 84 = 760$. Divide by 10 scores: 76. The group sizes determine each mean's weight.



Use errors to choose the next step

Identify the first mismatch

Compare the student's representation with the quantities or stated condition. Separate reading the situation, choosing a method, calculation, and interpretation.

Respond precisely

If the model is wrong, use a simpler example before repeating the procedure. If the method is sound and arithmetic is wrong, correct that step and verify the original result.

Review the exit ticket

For each item: 2 points for an accurate result with coherent reasoning; 1 for a sound method with a minor error or an unexplained correct answer; 0 for unsupported or missing reasoning. Use comments to select a next action. These are local instructional criteria.

AIG

Compare two methods, construct a counterexample, or create a new problem with a prescribed solution and justify its restrictions.

504 / EC

Follow the individual plan. Offer manageable item groups, documented time or format supports, a worked step, or an alternate response method while retaining the mathematical objective.

ESL

Preview terms and symbols in the questions. Use home-language rehearsal or a bilingual glossary, then ask for a concise explanation such as "The condition means ___, so I used ___."

Sources, preparation, and license

Objective: Compute a combined mean using group sizes. State alignment: not claimed. This is a focused college-entrance preparation resource, not a state course or a claim of course equivalency.

Official exam scope consulted September 15, 2026: College Board, Types of Math Tested, <https://satsuite.collegeboard.org/sat/whats-on-the-test/math/types> ; ACT, Mathematics Test Description, <https://www.act.org/content/act/en/products-and-services/the-act/test-preparation/description-of-math-test.html> .

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